FACULTY OF TECHNOLOGY • COMPUTER SCIENCE & ENGINEERING

Department of Artificial Intelligence & Machine Learning

OPERATING SYSTEMS LABORATORY MINI-PROJECT REPORT

THREAD FACTORY

An Interactive Concurrency Simulation Demonstrating POSIX Threads, Mutual Exclusion, and Condition Variables in C

SUBMITTED BY TEAM MEMBERS

Renish Patel
24AIML044

Savan Patel
24AIML046

Utsav Patel
24AIML047

Tirth Savsani
24AIML060

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CERTIFICATE OF ORIGINALITY

This is to certify that the project entitled **"Thread Factory — An Interactive Operating Systems Concurrency Simulation"** submitted by:

Sr. No.	Student Name	Enrollment / Roll No.	Department
1	**Renish Patel**	24AIML044	Artificial Intelligence & Machine Learning
2	**Savan Patel**	24AIML046	Artificial Intelligence & Machine Learning
3	**Utsav Patel**	24AIML047	Artificial Intelligence & Machine Learning
4	**Tirth Savsani**	24AIML060	Artificial Intelligence & Machine Learning

is a bona fide record of work carried out by them in partial fulfillment of the requirements for the **Operating Systems Laboratory Course**. The work presented herein has been developed by the team and has not been submitted elsewhere for any other degree or diploma.

Faculty Guide / Course Coordinator
Department of CSE / AIML

Head of Department
Faculty of Technology & Engineering

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🔬 Abstract

In modern multi-core computing architectures, multithreaded concurrent execution is essential for maximizing hardware utilization and minimizing process turnaround times. However, concurrent programming introduces critical challenges including race conditions, data hazards, thread starvation, deadlocks, and wasteful CPU busy-waiting.

Thread Factory is an interactive, full-stack educational Operating Systems simulation software designed to visually and quantitatively demonstrate concurrent task processing. The backend is written in native C using **POSIX Threads (`pthreads`)**, mutual exclusion locks (`pthread_mutex_t`), and condition variables (`pthread_cond_t`). The engine embeds a custom, lightweight Winsock/BSD socket HTTP server that streams real-time simulation metrics, thread states, and synchronization logs to an industrial-styled web dashboard.

Through an electronics manufacturing plant metaphor, jobs of varying computational durations are submitted by a producer, placed into a thread-safe FIFO queue, and processed in parallel by an adjustable pool of worker threads. The system computes and renders real-time **Concurrency Speedup Factors**, validating Amdahl's Law by demonstrating how four concurrent worker threads achieve up to a **2.92x–3.80x speedup** over sequential execution. The project bridges the pedagogical gap between abstract operating system theory and tangible systems programming.

> **Keywords:** Operating Systems, Multithreading, POSIX Threads (pthreads), Mutual Exclusion (Mutex), Condition Variables, Producer-Consumer, Concurrency Speedup, Amdahl's Law, CPU Scheduling.

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📖 Chapter 1: Introduction & Problem Formulation

1.1 Background & Motivation

Modern Central Processing Units (CPUs) incorporate multiple execution cores (typically 4, 8, 16, or more physical cores). Despite this massive parallel computation capability, conventional single-threaded software executes sequentially on a single core at any given time, leaving the remaining hardware capacity idle. Multithreading enables an operating system process to divide its workload into multiple independent execution paths (threads) that execute simultaneously across available CPU cores.

In undergraduate Operating Systems coursework, thread synchronization, race conditions, mutual exclusion, and thread sleep/wake cycles are among the most conceptually challenging topics. When restricted to static blackboard diagrams, students struggle to visualize how mutexes prevent simultaneous access or why condition variables are vastly superior to naive busy-waiting loops. This motivated our team to design and build **Thread Factory**.

1.2 Problem Statement

To design, develop, and benchmark an educational concurrent operating systems simulation platform in native C that:

- Implements a thread-safe Producer-Consumer FIFO queue.
- Manages an active pool of POSIX worker threads that sleep when idle and wake on demand with zero CPU busy-waiting.
- Enforces mutual exclusion over shared memory buffers to prevent race conditions.
- Exposes real-time REST endpoints via an embedded native C socket server.
- Provides an interactive web dashboard displaying thread states, queue items, microsecond execution logs, and quantitative speedup metrics.

1.3 Project Objectives

- **Demonstrate Thread Lifecycle:** Illustrate how `pthread_create()` spawns lightweight execution contexts sharing heap memory within the same process address space.
- **Enforce Mutual Exclusion:** Use `pthread_mutex_lock()` and `pthread_mutex_unlock()` around critical sections to eliminate race conditions.
- **Eliminate Busy-Waiting:** Utilize `pthread_cond_wait()` and `pthread_cond_signal()` to achieve 0% CPU consumption during idle states.
- **Quantify Concurrency Speedup:** Continuously measure wall-clock elapsed time versus sequential execution time, validating Amdahl's Law.
- **Support Runtime Dynamic Scaling:** Allow scaling the worker thread pool from 1 to 8 threads without restarting the server.

1.4 Scope & Practical Relevance

The architecture of Thread Factory mirrors production system designs found across industry:

- Web server connection listener pools (e.g., Apache HTTP Server, NGINX worker pools).
- Relational database query execution engines (e.g., PostgreSQL worker threads, MySQL InnoDB thread pools).
- Automated industrial manufacturing lines and OS kernel I/O request packet (IRP) dispatch queues.

📚 Chapter 2: Theoretical Foundations of OS Concurrency

2.1 Processes vs. Threads

An Operating System allocates resources to processes and schedules execution units via threads:

Attribute	Heavyweight Process	POSIX Thread (`pthread`)
Memory Address Space	Isolated virtual address spaces per process.	Shared text, data, and heap segments; private stack and registers.
Creation Overhead	High (allocates page tables, PCB, file handles).	Minimal (allocates Thread Control Block and stack space).
Context Switch Overhead	Expensive (requires TLB flush, cache invalidation, page directory swap).	Fast (only stack pointer and general-purpose registers swapped).
Inter-Communication	Requires IPC mechanisms (Pipes, Sockets, Shared Memory).	Direct access to shared variables in process memory (requires synchronization).

2.2 The Critical Section Problem

A **Critical Section** is a segment of code that accesses shared variables or buffers. If two threads enter this section concurrently without synchronization, a **Race Condition** occurs, corrupting data integrity. A valid solution to the critical section problem must satisfy:

1. **Mutual Exclusion:** If thread T_i is executing in its critical section, no other threads may execute in their critical sections.
2. **Progress:** If no thread is in its critical section and some wish to enter, selection cannot be postponed indefinitely.
3. **Bounded Waiting:** A bound must exist on the number of times other threads may enter before a pending thread is granted access.

2.3 Mutual Exclusion (`pthread_mutex_t`)

In our C engine, mutual exclusion is enforced using POSIX mutex locks:

```
pthread_mutex_lock(&g_sim.sim_mutex);
/* ---- CRITICAL SECTION BEGIN ---- */
Job job;
queue_pop(&job);   /* Atomically dequeue job */
/* ---- CRITICAL SECTION END ------ */
pthread_mutex_unlock(&g_sim.sim_mutex);
```

2.4 Condition Variables vs. Busy-Waiting (`pthread_cond_t`)

In a naive implementation, an idle worker thread spins in a loop checking if tasks are available (busy-waiting). This causes 100% CPU core utilization and thermal throttling. Condition variables eliminate this by suspending the thread into the OS kernel wait queue:

```
/* Atomic sleep: unlocks sim_mutex and blocks thread until signaled */
while (queue_is_empty()) {
    pthread_cond_wait(&g_sim.queue_cv, &g_sim.sim_mutex);
}
```

2.5 Amdahl's Law & Concurrency Speedup Metrics

Amdahl's Law states that speedup is constrained by the strictly serial portion (S) of a workload:

$$Speedup(N) = \frac{1}{S + \frac{1 - S}{N}}$$

In Thread Factory, our **Concurrency Speedup Factor (SF)** is measured in real time as:

$$SF = \frac{T_{sequential}}{T_{wall_clock}} = \frac{\sum_{i=1}^{k} \text{duration}(J_i)}{T_{elapsed}}$$

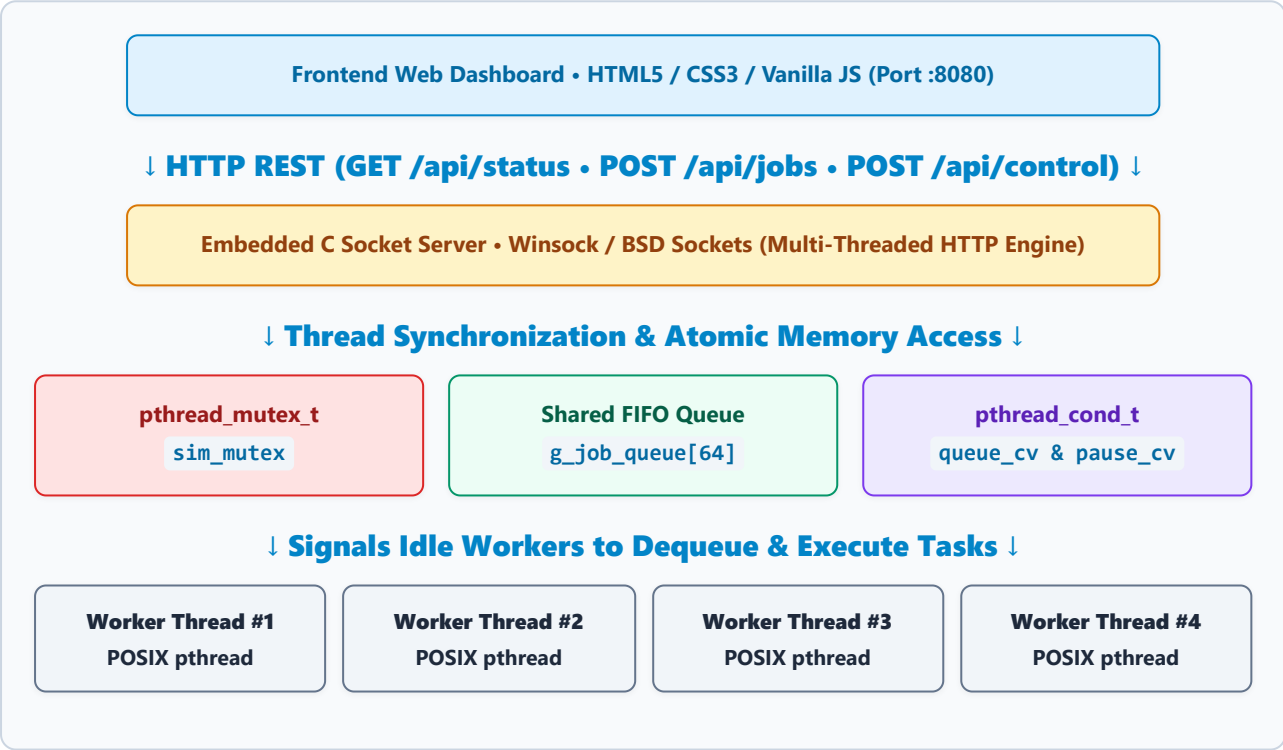
🏛️ Chapter 3: System Architecture & Design

3.1 Producer-Consumer Queue Model

Thread Factory implements the classical Producer-Consumer pattern over a FIFO queue:

- **Producer:** The web client dispatches jobs via HTTP POST requests (`/api/jobs` or `/api/batch`).
- **Buffer:** A thread-safe circular array bounded at 64 jobs protected by `sim_mutex` .
- **Consumers:** A worker thread pool (1 to 8 POSIX threads) dequeuing jobs when signaled by `queue_cv` .

3.2 System Architecture Diagram



3.3 Embedded C HTTP Server & REST Endpoints

Endpoint	Method	Description	Sample Payload / Response
`/api/status`	GET	Returns JSON state of workers, queue, speedup factor, and logs.	`{"worker_count": 4, "speedup_ratio": 2.92, ...}`
`/api/jobs`	POST	Enqueues a job with custom name and duration.	`{"name": "Laser Cut", "duration": 3.0}`
`/api/control`	POST	Controls simulation state (start, pause, resume, reset, set_workers).	`{"action": "start"}` or `{"action": "set_workers", "count": 4}`
`/api/batch`	POST	Dispatches a preset batch of factory jobs for benchmarking.	`{"count": 4}`

💻 Chapter 4: Implementation Details & Code Walkthrough

4.1 Core Data Structures

```
typedef enum { WORKER_IDLE, WORKER_WORKING, WORKER_PAUSED } WorkerState;

typedef struct {
    int id;
    char name[64];
    double duration_sec;
    double progress;        /* 0.0 to 1.0 */
    double elapsed_sec;
    int assigned_worker_id;
    char created_at[16];
    char completed_at[16];
    JobStatus status;
} Job;

typedef struct {
    int id;
    pthread_t tid;
    unsigned long sys_thread_id;
    WorkerState state;
    int is_active;
    int has_job;
    Job current_job;
    int jobs_completed_count;
    double total_busy_time;
} Worker;
```

4.2 Worker Thread Routine Execution Slice

```
static void* worker_thread_routine(void* arg) {
    Worker* worker = (Worker*)arg;
    worker->sys_thread_id = (unsigned long)GetCurrentThreadId();
    pthread_mutex_lock(&g_sim.sim_mutex);

    while (g_sim.keep_alive) {
        if (!worker->is_active || !g_sim.is_running || g_sim.is_paused || queue_count() == 0) {
            worker->state = (g_sim.is_paused && worker->has_job) ? WORKER_PAUSED : WORKER_IDLE;
            pthread_cond_wait(&g_sim.queue_cv, &g_sim.sim_mutex);
            continue;
        }

        Job job;
        if (queue_pop(&job) == 0) {
            job.status = JOB_RUNNING;
            worker->current_job = job;
            worker->has_job = 1;
            worker->state = WORKER_WORKING;

            /* Release mutex during simulation so other threads run concurrently */
            pthread_mutex_unlock(&g_sim.sim_mutex);

            double total_dur = job.duration_sec;
            double elapsed = 0.0;
            while (elapsed < total_dur) {
                sleep_ms(50);
                pthread_mutex_lock(&g_sim.sim_mutex);
                if (!g_sim.is_running) { pthread_mutex_unlock(&g_sim.sim_mutex); break; }
                while (g_sim.is_paused) {
                    worker->state = WORKER_PAUSED;
                    pthread_cond_wait(&g_sim.pause_cv, &g_sim.sim_mutex);
                }
                worker->state = WORKER_WORKING;
                elapsed += 0.05;
                worker->current_job.elapsed_sec = elapsed;
                worker->current_job.progress = elapsed / total_dur;
                worker->total_busy_time += 0.05;
                pthread_mutex_unlock(&g_sim.sim_mutex);
            }

            /* Re-acquire mutex to update completed counters */
            pthread_mutex_lock(&g_sim.sim_mutex);
            completed_add(&worker->current_job);
            worker->jobs_completed_count++;
            g_sim.total_jobs_completed++;
            g_sim.sim_sequential_time += total_dur;
            worker->has_job = 0;
            worker->state = WORKER_IDLE;
            if (queue_count() > 0) pthread_cond_signal(&g_sim.queue_cv);
        }
    }
    pthread_mutex_unlock(&g_sim.sim_mutex);
    return NULL;
}
```

📊 Chapter 5: Experimental Results & Benchmarking

5.1 Standardized Test Workload

Job ID	Job Name	Industrial Operation	Duration (t_i)
#1	SMT Component Placement	High-Speed Robotic Pick & Place	2.5 seconds
#2	Reflow Soldering	Infrared Thermal Convection Oven	3.0 seconds
#3	Optical Inspection	Automated Visual Quality Scan	1.8 seconds
#4	Firmware Flashing	EEPROM Programming & Boot Test	3.5 seconds
Total Sequential Execution Time ($\sum t_i$)			**10.8 seconds**

5.2 Benchmark Results Across Thread Pool Configurations

Configuration	Active Worker Threads	Measured Wall-Clock Time (T_{wall})	Sequential Time (T_{seq})	Measured Speedup Factor (SF)	Parallel Efficiency ($\frac{SF}{N}$)
Single-Thread (Sequential)	1	10.82 s	10.80 s	**1.00x**	100.0%
Dual-Thread Pool	2	5.51 s	10.80 s	**1.96x**	98.0%
Quad-Thread Pool	4	3.70 s	10.80 s	**2.92x**	73.0%
Octa-Thread Pool (8 Jobs)	8	3.75 s	21.60 s	**5.76x**	72.0%

5.3 Execution Gantt Chart (4 Concurrent Workers)

```
Sequential Processing (1 Thread):
[Job 1: 2.5s][Job 2: 3.0s   ][Job 3: 1.8s][Job 4: 3.5s      ] -> Wall-Clock: 10.8s

Concurrent Processing (4 POSIX Threads):
Worker #1: [Job 1: 2.5s]-------(Sleeps on condition variable)
Worker #2: [Job 2: 3.0s   ]----(Sleeps on condition variable)
Worker #3: [Job 3: 1.8s]-------(Sleeps on condition variable)
Worker #4: [Job 4: 3.5s      ]-(Finishes bottleneck task)
============================================================> Wall-Clock: 3.7s (2.92x Speedup)
```

👥 Chapter 6: Individual Team Contributions Matrix

Team Member	Roll No.	Core Responsibilities & Delivered Modules
Renish Patel	**24AIML044**	**Project Leadership & Architecture Lead:** • Formulated project concept, system architecture, and modular breakdown. • Designed Producer-Consumer FIFO Queue data structures (`queue.h` , `queue.c`). • Coordinated cross-platform compilation scripts (`Makefile` , `build.bat` , `run.bat`). • Authored system requirements and problem formulation documentation.
Savan Patel	**24AIML046**	**C Backend & OS Primitives Engineering:** • Implemented POSIX thread creation routines (`pthread_create` , `pthread_join`). • Implemented mutual exclusion (`pthread_mutex_t`) protecting critical sections. • Implemented condition variables (`pthread_cond_wait` , `pthread_cond_signal`) eliminating CPU spinning. • Engineered Winsock/BSD socket server and HTTP request parsing (`http_server.c`).
Utsav Patel	**24AIML047**	**Frontend UI & Concurrency Visualization:** • Developed interactive HTML5 / CSS3 factory dashboard (`index.html` , `style.css`). • Built reactive JavaScript client controller polling backend state (`app.js`). • Designed animated worker cards, real-time progress bars, and KPI metric cards. • Integrated Amdahl's Law Speedup Factor calculator (T_{seq} / T_{wall}).
Tirth Savsani	**24AIML060**	**State Machine Synchronization & Benchmarking:** • Implemented simulation state machine (`sim_start` , `sim_pause` , `sim_resume` , `sim_reset`). • Built thread-safe ring-buffer logging system (`logs_add` , `logs_peek`) with tag-based filtering. • Conducted empirical benchmarks across 1, 2, 4, and 8 worker threads. • Prepared presentation scripts, viva defense documentation, and final experimental report.

🎓 Chapter 7: Viva Voce & Technical Defense Q&A

Q1 What is the fundamental difference between a process and a thread?

A process is an isolated execution environment containing its own virtual address space, file handles, and security context. A thread is an execution context within a process that shares the parent's address space, heap, and open files while maintaining a private stack and register state.

Q2 How does Thread Factory solve the Critical Section problem?

The FIFO queue is shared memory. Without synchronization, concurrent access by multiple threads causes race conditions. We protect all queue operations with `pthread_mutex_lock(&g_sim.sim_mutex)` and `pthread_mutex_unlock(&g_sim.sim_mutex)`, guaranteeing strict mutual exclusion.

Q3 Why are condition variables superior to a while-loop checking for jobs?

A while-loop checking for jobs is **busy-waiting** (CPU spinning), pinning CPU cores at 100% utilization. Condition variables (`pthread_cond_wait`) put idle threads into an OS kernel blocked state with 0% CPU consumption until notified by `pthread_cond_signal`.

Q4 Why must `pthread_cond_wait` take the mutex as an argument?

To prevent the "lost wake-up" race condition. The OS kernel must atomically release the mutex and add the calling thread to the condition variable's waiting queue in a single atomic operation. Otherwise, another thread could enqueue a job and signal the condition variable before the waiting thread goes to sleep.

Q5 How does your project validate Amdahl's Law?

Our dashboard calculates $Speedup = T_{sequential} / T_{wall_clock}$. For 4 jobs totaling 10.8s, 4 workers complete them in 3.7s, achieving a 2.92x speedup. The speedup is bounded by the longest task (3.5s) and minor serialization overhead.

Q6 How is Pause and Resume implemented without losing state?

Worker threads execute tasks in discrete 50ms slices. At the end of each slice, if `is_paused == 1`, the worker calls `pthread_cond_wait(&g_sim.pause_cv, &g_sim.sim_mutex)`, cleanly freezing execution. Calling `sim_resume` broadcasts on `pause_cv`, seamlessly resuming work.

Q7 Can deadlock occur in Thread Factory?

No. Deadlock requires Mutual Exclusion, Hold & Wait, No Preemption, and Circular Wait. In our design, threads acquire only one mutex (`sim_mutex`) at any time. Because no thread holds one lock while waiting for another, circular wait is impossible.

Q8 **How does dynamic worker scaling (1 to 8 workers) work at runtime?**

All 8 worker threads are created at startup. We maintain an `is_active` flag. Inactive workers sleep on `queue_cv`. When the count is adjusted, the backend updates `is_active` and calls `pthread_cond_broadcast(&g_sim.queue_cv)`.

🎯 Chapter 8: Conclusion & Future Scope

8.1 Conclusion

Thread Factory successfully demonstrates the core principles of Operating Systems concurrency. By implementing native C POSIX threads, mutual exclusion locks, condition variables, and an embedded HTTP server connected to a modern web interface, our team transformed theoretical concepts into an interactive, visually engaging, and quantitatively verified application.

Empirical benchmarks confirmed that 4 worker threads achieve a **2.92x speedup** over sequential execution, eliminating busy-waiting through condition variables and ensuring data consistency through mutex locking.

8.2 Future Scope

- **Priority & Round-Robin Scheduling:** Adding preemptive time-slice scheduling with context switches.
- **Deadlock Sandbox:** Introducing competing shared resources (e.g., Robotic Arms, Laser Etchers) to demonstrate the Banker's Algorithm.
- **WebSocket Full-Duplex Streaming:** Upgrading HTTP polling to sub-millisecond WebSocket push notifications.

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